



ရန်ကုန်အရှေ့ပိုင်းတက္ကသိုလ်၊ ၂၀၂၆ ပညာသင်နှစ်၊ စာစဉ်အဖြေလွှာစာအုပ်

အမည်

ရုံးအမှတ်

အထူးပြု

Module No./Module Name

Assignment I / II

- မောင်ဒေဝရှင်စိုး

- ၃-၇၁-၁၆ (၁၇၆)

- ရွှေမင်း

- Bot - 2101 / (Angiosperms)

- I

1. Differences between classical and modern taxonomy

Classical Taxonomy	Modern Taxonomy
It is called old systematics or Alpha (α) taxonomy or Taxonomy	It is called Neosystematics or Biosystematics or Omega (Ω) taxonomy
It is pre Darwinean	It is post Darwinean
Species is considered as basic unit and is static	Species is considered as dynamic and ever changing
Classification is mainly based on morphological characters	Classification is mainly based on morphological, reproductive characters and phylogenetic (evolutionary) relationship of the organism
This system is based on the observation of a few samples/individuals	This system is based on the observation of large number of samples/individuals

2. Principles of Nomenclature

There are six principles -

- (1) Botanical nomenclature is independent of zoological nomenclature. The code applies equally to names of taxonomic groups treated as plants whether or not these groups were originally so treated (Plants do not include Bacteria).
- (2) Application of names of taxonomic groups is determined by means of nomenclature types.
- (3) The nomenclature of a taxonomic group is based upon priority of publication.
- (4) Each taxonomic group with a particular circumscription, position, and rank can bear only one correct name, the earliest that is in accordance with the rules, except in specific cases.

- 5) Scientific names of taxonomic groups are treated as Latin regardless of their derivation.
- (6) The rules of nomenclature are retroactive ~~unless~~^{unless} expressly limited.
3. Family Descriptions: The Nymphaeaceae consist of aquatic, annual or perennial herbs, with a milky latex often present. The underground stems are rhizomatous or tuberous. The stem vasculature is an atactostele or eustele. The leaves are simple, often peltate, stipulate or exstipulate, floating, spiral, usually orbicular in shape. The inflorescence consists of a solitary, floating or emergent flower. Flowers are bisexual, actinomorphic, and hypogynous or epigynous, with long peduncles arising from the underground stem. The perianth is usually differentiated into calyx and corolla, the parts spirally arranged. The calyx consists of 4-6 [up to 14], aposepalous sepals. The corolla consists of 8-many [0], apopetalous petals, the inner of which grade into laminar stamens. Stamens are numerous ~~in~~ spiral, apopetalous petals, the inner of which grade apostemonous; the filaments are laminar to the outside, grading morphologically into petals, to terete toward the flower center; anthers are longitudinal in dehiscence, dithecal, with thecae and connective often extending beyond the anther. The gynostemium is syncarpous, with a superior or inferior ovary, and 3-many ~~carpels~~^{carpels}, ~~apend~~, placentation is ~~axile~~^{axile} lamellate or parietal; ovules are anatropous, ~~bitegmic~~^{bitegmic}, and numerous per carpel. The fruit is a berry.



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အမည်

- မောင်အောင်(ရ.ပိုင်)

ရံအမှတ်

- ၃-၇၃-၁၆ (၀၅၆)

အထူးပြု

- ဂုဏ်ဗေဒ

Module No./Module Name

- Bot. 2101 / (Angiosperms)

Assignment I / II

- II

1. Solanaceae - Nightshade family (Latin for sleeping or comforter, after narcotic properties of some), 94 genera / 2950 species.

Family Descriptions: The Solanaceae consist of herbs, shrubs, trees, or lianas, with prickles present in some taxa, many with stellate trichomes. The leaves are simple, pinnate, or ternate, usually spiral and exstipulate. The inflorescence is of solitary flowers or cyme units. The flowers are bisexual, actinomorphic, rarely zygomorphic. The perianth is biseriate, dichlamydeous, usually tubular, rotate, or salverform, hypanthium absent. The calyx is synsepalous, persistent, sometimes accrescent, with 5 calyx lobes. The corolla is sympetalous and with 5 [4, 6] convolute, imbricate, or valvate lobes, with usually ~~involute~~ ^{involute} (plicate) aestivation. The stamens are 5 [rarely 4 or 2+2 staminodes], antisepalous and epipetalous, the anthers often connivent, with staminodes ~~rarely~~ ^{rarely} present. Anthers are longitudinal or poricidal in dehiscence. The gynoecium is syncarpous, with a superior ovary, 2 [rarely 3-5] carpels, and 2 [rarely 1 or 4-5] locules. Placentation is axile, rarely basal, ovules are variable in type, unitegmic, [rarely 1-few] per carpel. The fruit is a berry, drupe, or capsule (often septicidal). The seeds are endospermous. Alkaloids and internal phloem (inner to the xylem, surrounding pith) are present in many family members.

Classification and Distribution: The Solanaceae are currently classified into seven subfamilies (Olmstead et al. 1999). Members of the family have mostly worldwide distributions, concentrated in South America.

Economic Importance includes many edible plants, such as *Capsicum* (peppers), *Lycopersicon esculentum* (tomato), *Physalis philadelphica* (tomatillo), and *Solanum tuberosum* (potato), and the infamous fumatory *Nicotiana tabacum* (tobacco). Alkaloids from various taxa have medicinal properties (e.g., atropine from *Atropa belladonna*), hallucinogenic properties (e.g., *Datura*, Jimson weed), or are deadly poisons (e.g., *Datura*, *Solanum* spp.) or known carcinogens (e.g., *Nicotiana tabacum*); some are used as ornamental cultivars, others are noxious weeds.